EXPT NO:5	DEVELOP INTERACTIVE DASHBOARDS FOR DYNAMIC DATA EXPLORATION USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. **What is dynamic data exploration in BI tools, and why is it important?**

Dynamic data exploration is the process of interacting with data in real time using Business Intelligence (BI) tools such as Tableau and Power BI. Instead of viewing static reports, users can filter, sort, and analyze data from different perspectives.

It is important because it:

- Allows users to explore data based on their specific requirements.
- Helps identify trends, patterns, and anomalies quickly.
- Supports faster and more informed decision-making.
- Makes dashboards more flexible and user-friendly by enabling real-time analysis.

2. **What are slicers and filters, and how do they enhance dashboard interactivity?**

   Slicers and filters are interactive tools used to display only the data that matches selected conditions. They help users focus on relevant information without changing the original dataset.

   For example, a user can filter sales data by **region**, **product category**, or **date range**. Once a selection is made, all related charts and KPI cards update automatically. This makes dashboards easier to navigate, improves user experience, and enables quicker analysis of specific business scenarios.

3. **How do Tableau and Power BI support drill-down and drill-through features?**

   Both Tableau and Power BI provide features that allow users to explore data at different levels of detail. Drill-down enables users to move from summarized information to more detailed data, such as from Year → Quarter → Month → Day**.** Drill-through allows users to navigate to a separate report page containing detailed information related to a selected data point.

   These features make it easier to perform in-depth analysis without overcrowding the main dashboard and help users understand the reasons behind business trends.

4. **What is the role of parameters in creating interactive dashboards?**

   Parameters are user-defined controls that allow users to change the behavior of a dashboard dynamically. They provide flexibility by letting users switch between different measures, calculations, or visualization options without modifying the report design.

   For example, a parameter can allow users to switch between viewing **Sales** and **Profit** in the same chart. This makes dashboards more interactive, customizable, and useful for analyzing different business metrics.

5. **Name some advanced interactivity features in Tableau and Power BI.**

Both Tableau and Power BI provide several advanced features that improve dashboard interactivity and make data analysis more effective.
Some commonly used features include:

- Filters and Slicers for dynamic data selection.
- Drill-down and Drill-through for detailed analysis.
- Parameters to switch between different KPIs or views.
- Tooltips to display additional information when hovering over visuals.
- Bookmarks and Highlight Actions to create interactive navigation and storytelling.

These features enable users to explore data efficiently and gain deeper insights from interactive dashboards.

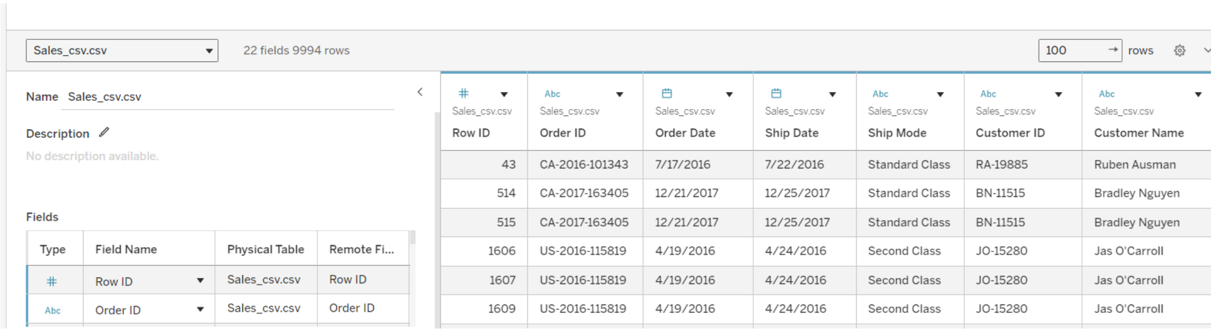
IN-LAB EXERCISE:

OBJECTIVE:

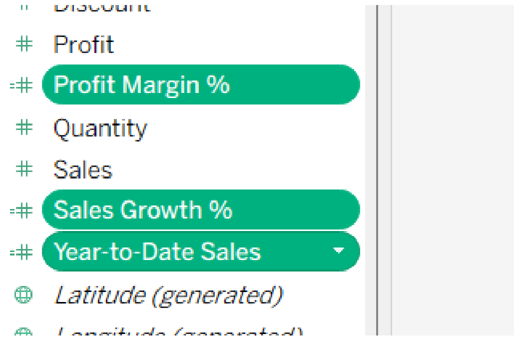
To design and develop interactive dashboards in Tableau and Power BI for dynamic data exploration, enabling users to analyze, filter, and drill down into data for deeper insights.

PROCEDURE:

1. **Data Collection:**
    - Import datasets from sources such as Excel, CSV, SQL Server, or online APIs.
    - Dataset: *Retail Sales Dataset* with fields such as Order Date, Region, Product Category, Sales, Profit, Quantity.
2. **Data Preparation and Transformation:**
    - Ensure proper date formats for time-based analysis.
    - Handle missing or incorrect values.

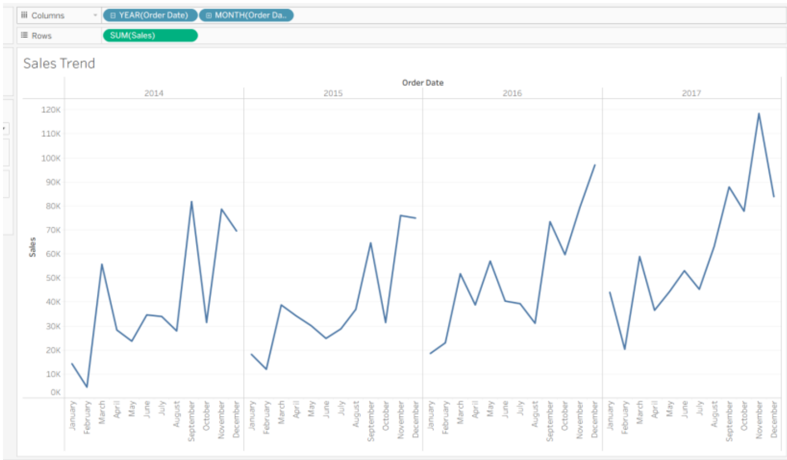


    - Create calculated fields (e.g., Profit Margin %, Sales Growth %, Year-to-Date Sales).



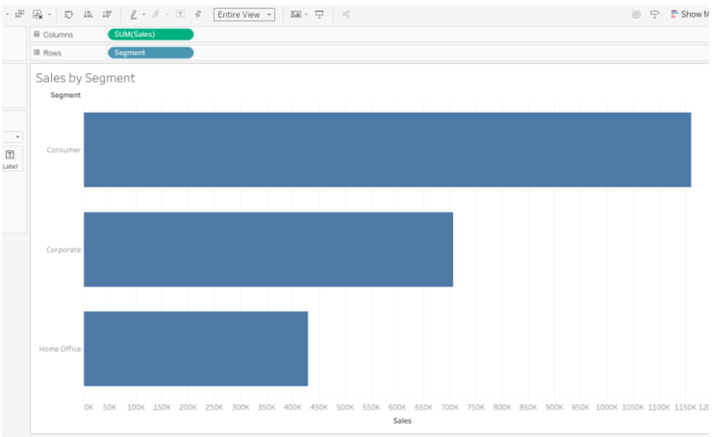
3. **Data Aggregation and Hierarchies:**
    - Create hierarchies (e.g., Year → Quarter → Month → Day).

Created a date hierarchy using the Order Date field, enabling drill-down analysis from Year → Quarter → Month → Day:



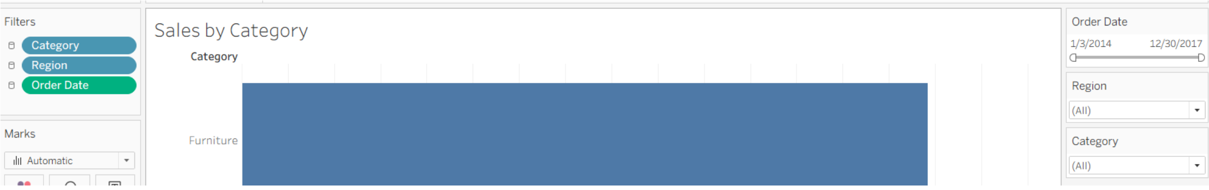
- Aggregate data by region, product category, or customer segment.

Demonstrating SUM:



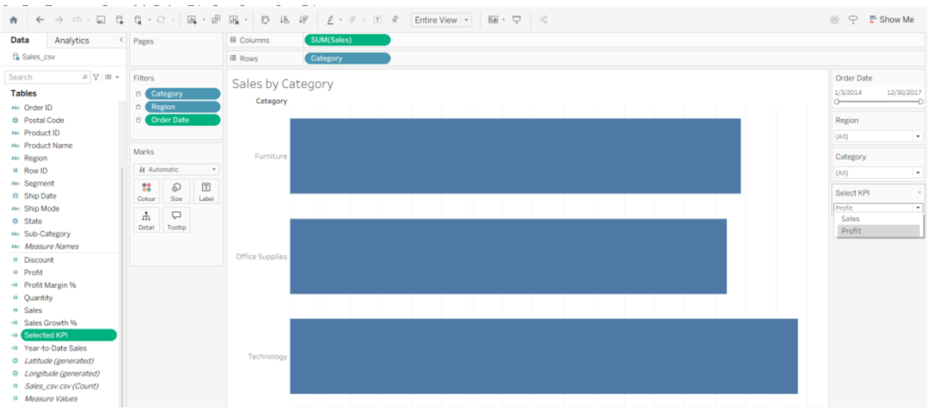
4. **Designing Interactive Elements:**

- **Filters & Slicers:** Allow filtering by product category, date range, or region.

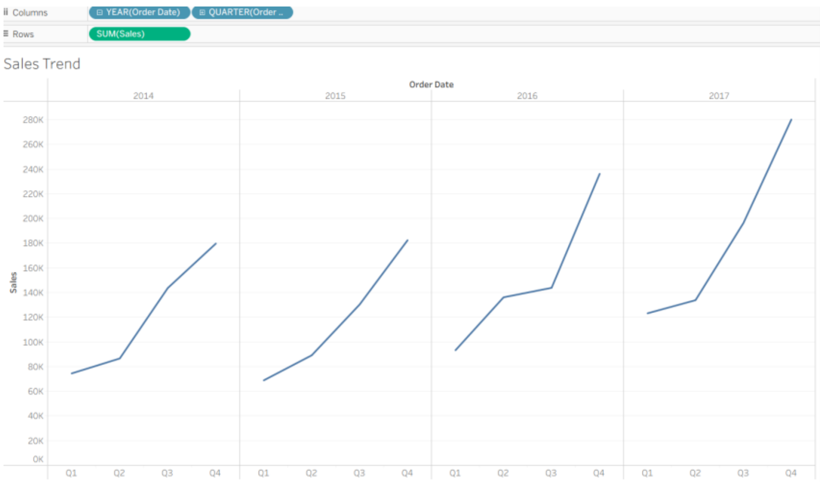


- **Parameters:** Let users switch between KPIs (e.g., Sales vs. Profit).

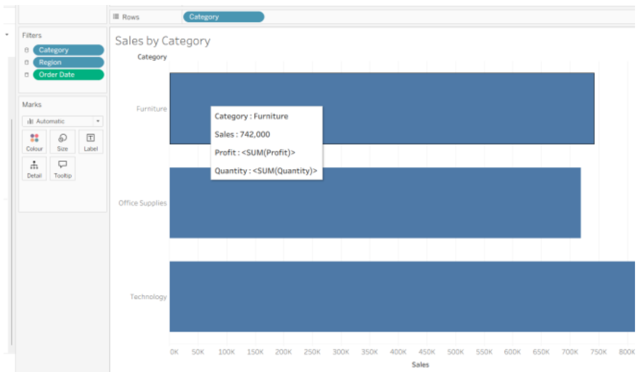
Created a parameter to allow users to dynamically switch between Sales and Profit within the same visualization:



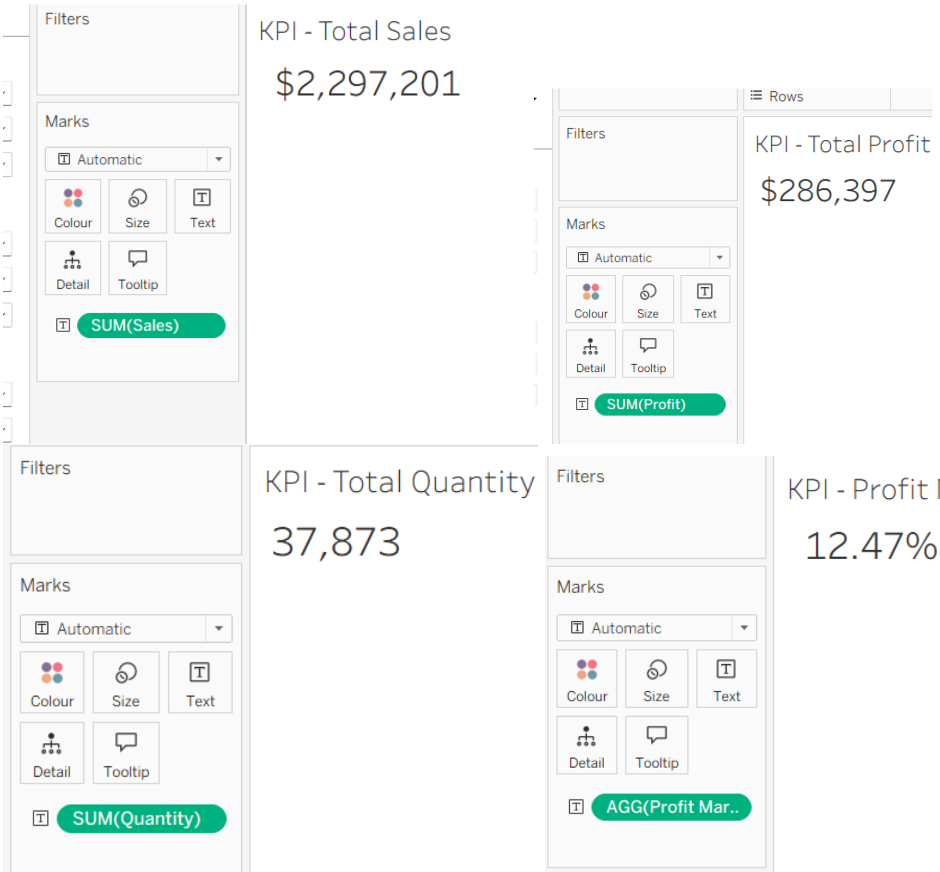
- **Drill-down:** Enable users to explore data from summary to detail.



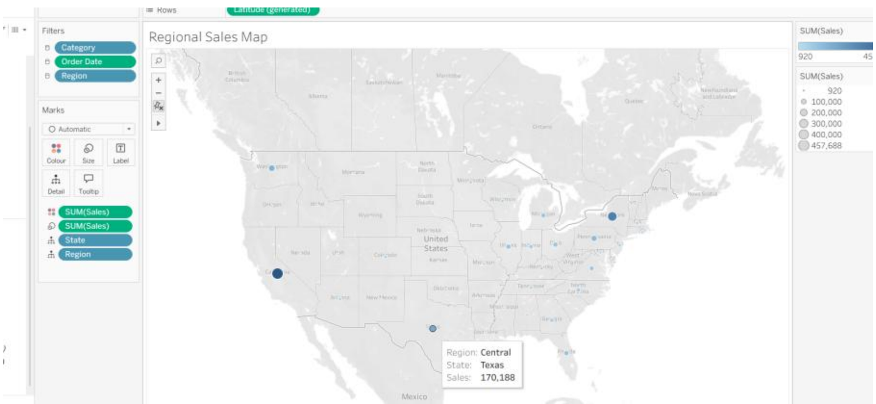
- **Tooltips:** Add contextual details when hovering over data points.

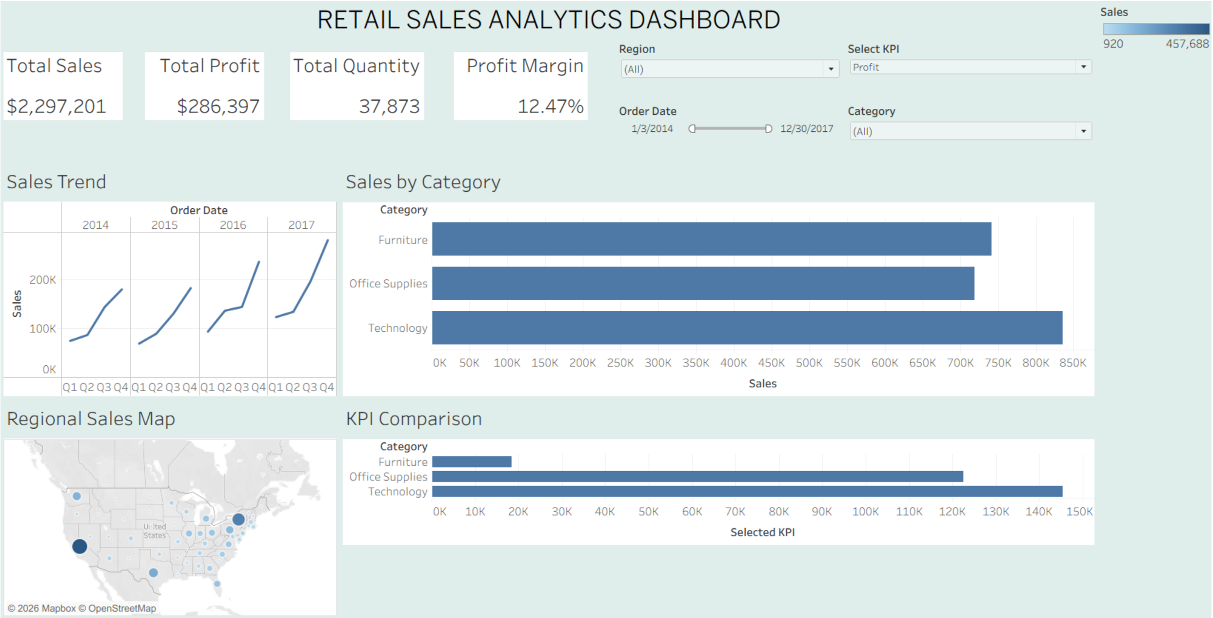


5. **Dashboard Development:**
    - **Tableau:** Use interactive filters, parameter controls, and highlight actions.
    - **Power BI:** Use slicers, bookmarks, and drill-through pages.
    - Add KPI cards for key metrics like *Total Sales*, *Total Profit*, *Top Selling Product.*



    - Use maps for regional sales analysis, line charts for trends, and bar charts for category comparisons.



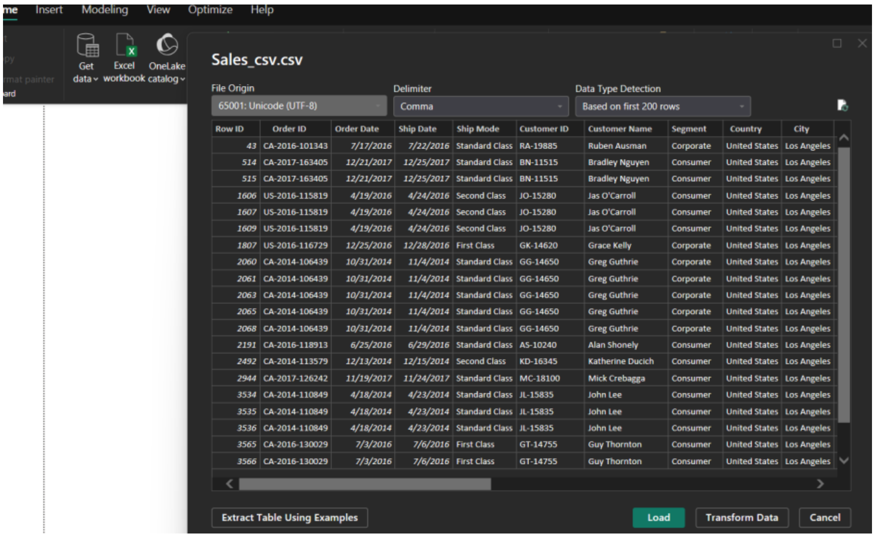


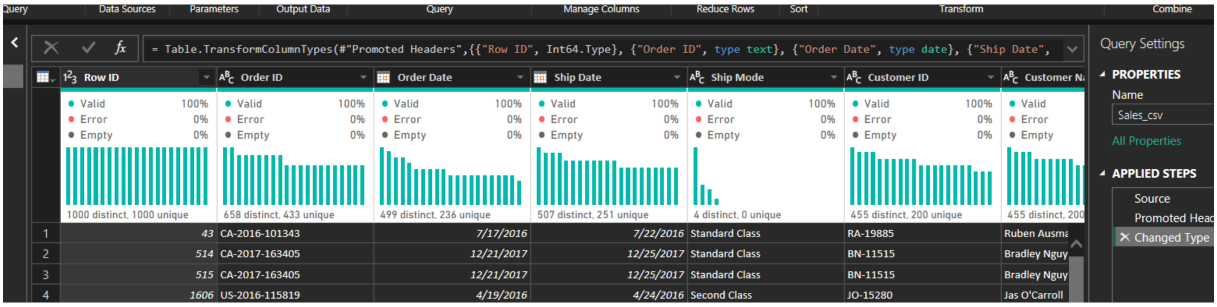
6. **Testing Interactivity:**
    - Ensure slicers and filters update all relevant visuals dynamically.
    - Test drill-down and drill-through functionality for accuracy.
7. **Publishing and Sharing:**

- **Tableau:** Publish on Tableau Public or Tableau Server.
- **Power BI:** Publish to Power BI Service and share via workspace access or public link.

In PowerBI:

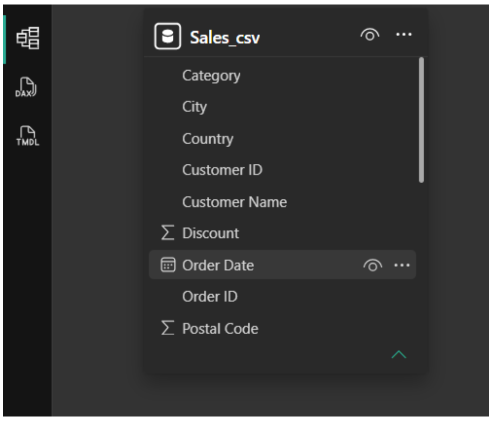
1. **Data Collection:**
    - Import datasets from sources such as Excel, CSV, SQL Server, or online APIs.
    - Dataset: *Retail Sales Dataset* with fields such as Order Date, Region, Product Category, Sales, Profit, Quantity.
2. **Data Preparation and Transformation:**
    - Ensure proper date formats for time-based analysis.
    - Handle missing or incorrect values.

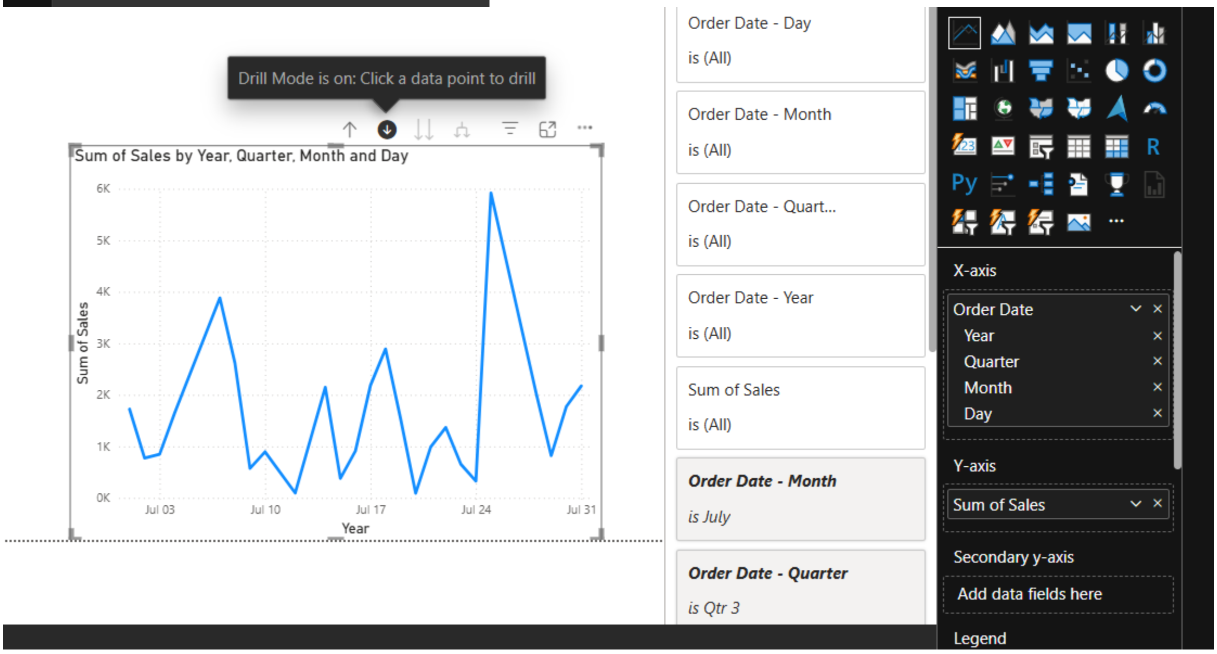




3. **Data Aggregation and Hierarchies:**
    - Create hierarchies (e.g., Year → Quarter → Month → Day).

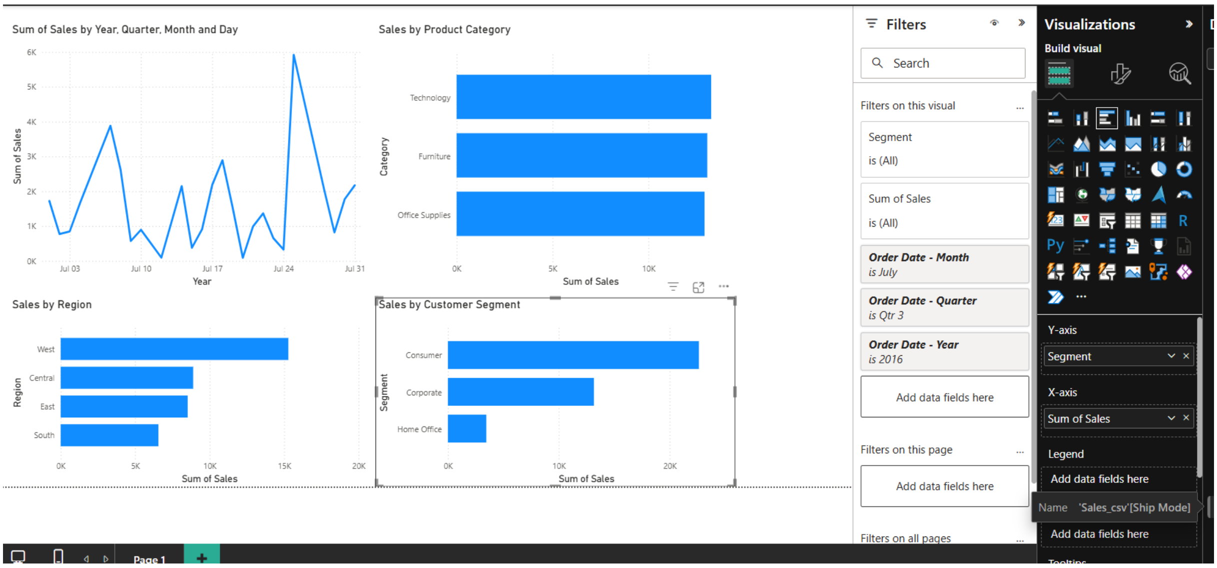
Created a date hierarchy using the Order Date field to enable drill-down analysis from Year to Quarter, Month, and Day:



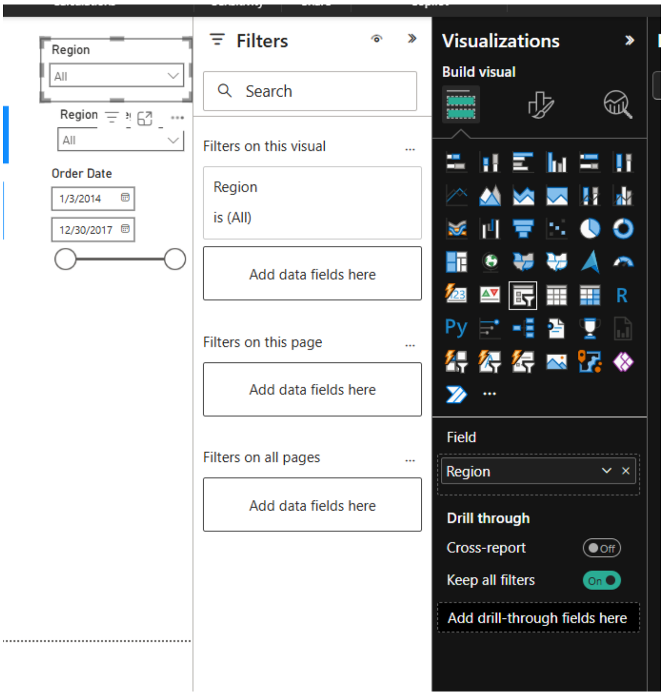


- Aggregate data by region, product category, or customer segment.

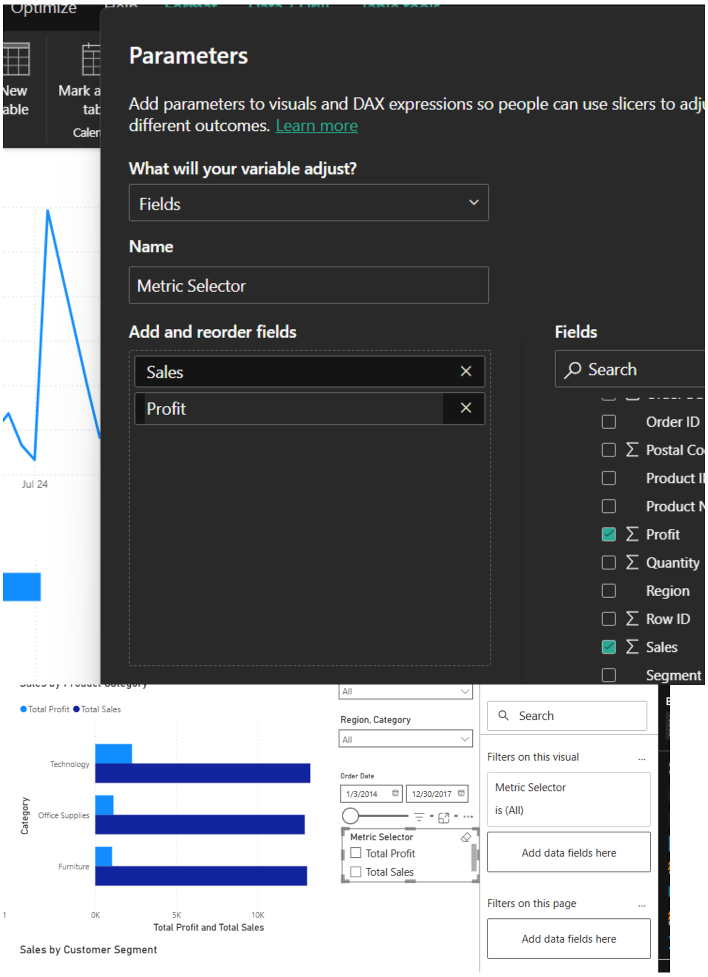
Demonstrating the SUM aggregation:

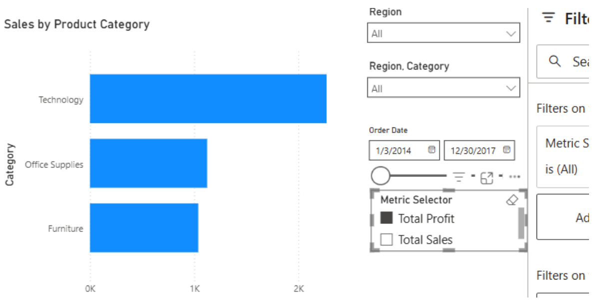


4. **Designing Interactive Elements:**
    - **Filters & Slicers:** Allow filtering by product category, date range, or region.

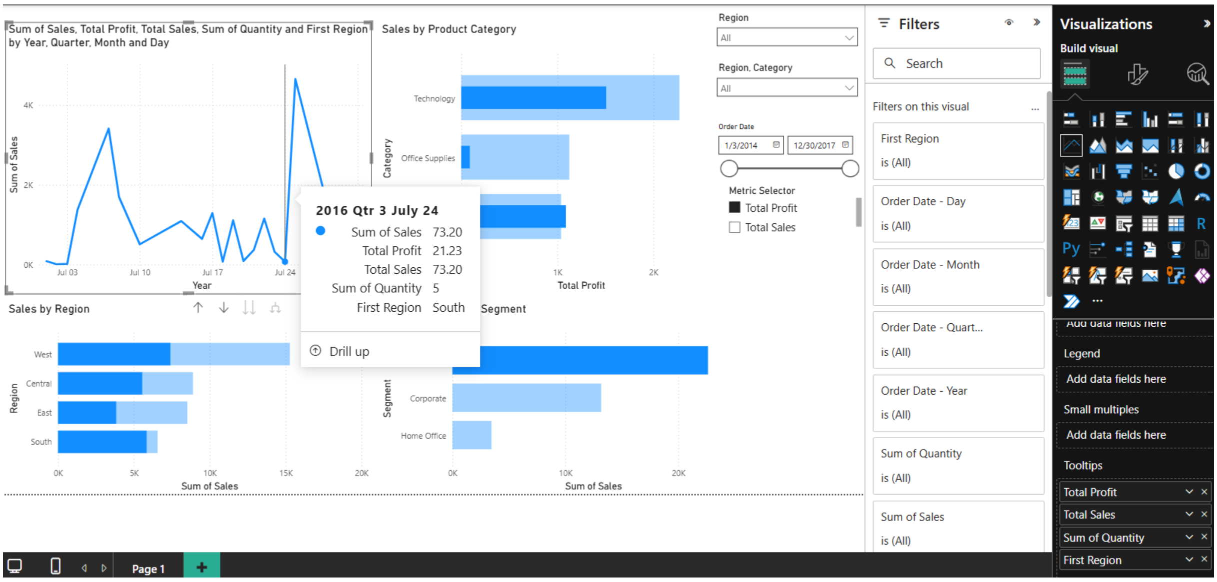


- **Parameters:** Let users switch between KPIs (e.g., Sales vs. Profit).



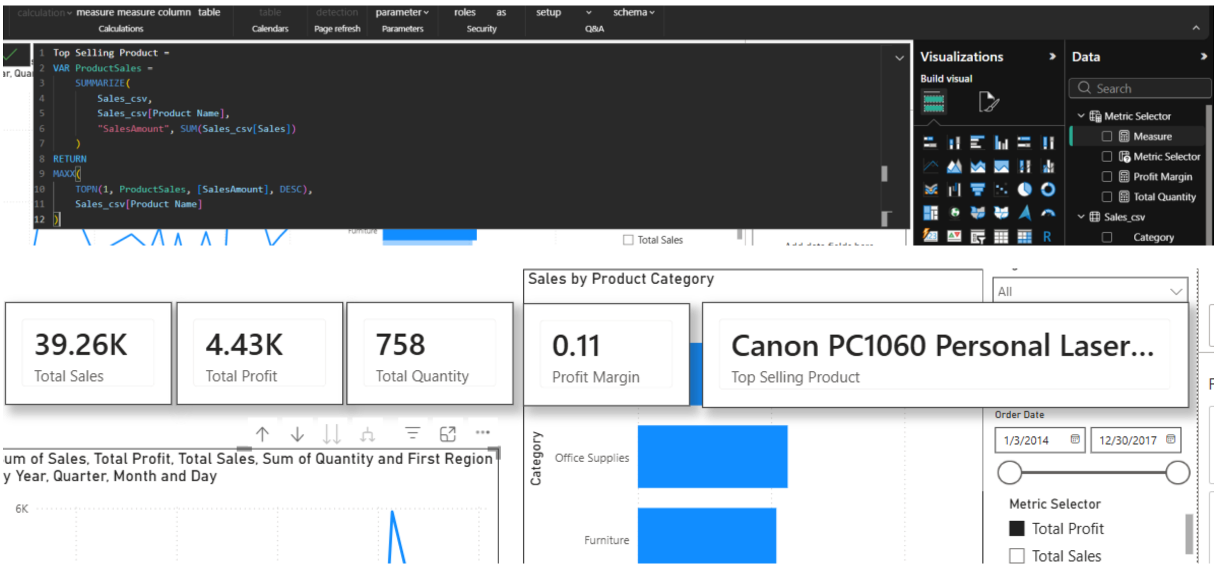


- **Tooltips:** Add contextual details when hovering over data points.



5. **Dashboard Development:**

   - **Tableau:** Use interactive filters, parameter controls, and highlight actions.
   - **Power BI:** Use slicers, bookmarks, and drill-through pages.
   - Add KPI cards for key metrics like *Total Sales*, *Total Profit*, *Top Selling Product*.



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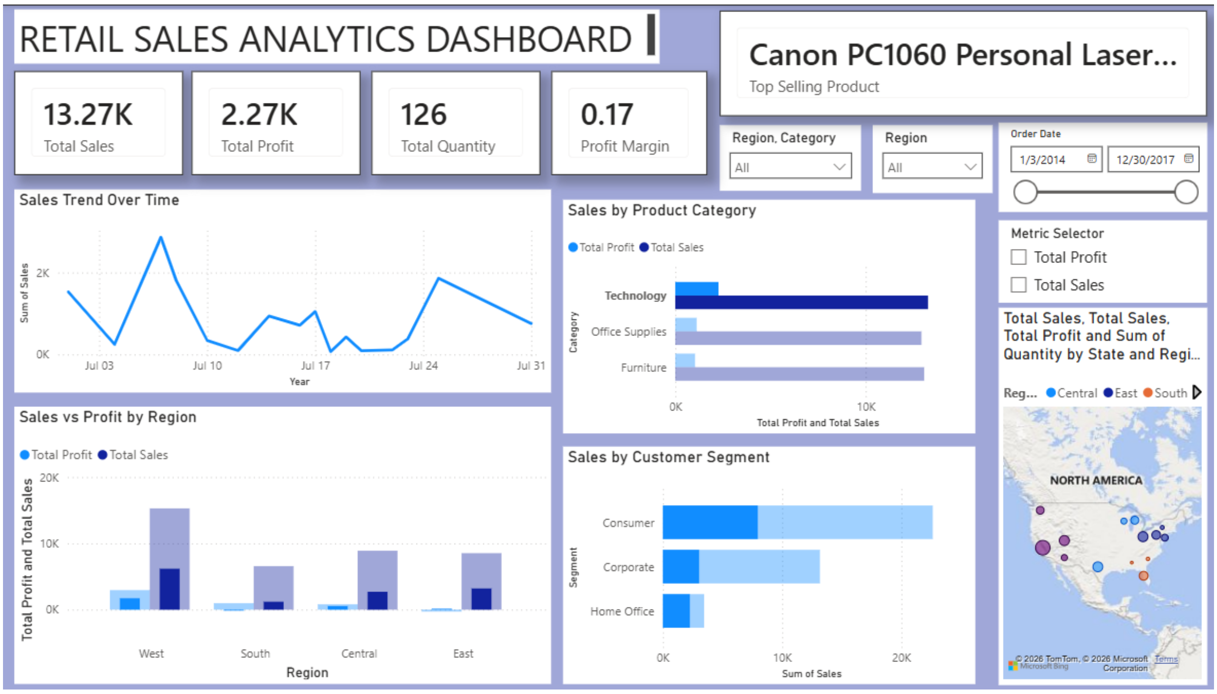

6. **Testing Interactivity:**
   - Ensure slicers and filters update all relevant visuals dynamically.
   - Test drill-down and drill-through functionality for accuracy.
7. **Publishing and Sharing:**

- **Tableau:** Publish on Tableau Public or Tableau Server.
- **Power BI:** Publish to Power BI Service and share via workspace access or public link.

Published link:https://app.powerbi.com/groups/me/reports/ff7cb11d-8a42-47e7-ae33-1c844ed42832/678c9b74778ece967cbd?experience=power-bi

Experiment Dataset:

Retail Sales Dataset

Fields Used:

- **Order Date – Date of the order**
- **Region – Sales region**
- **Product Category – Category of product sold**
- **Sales – Sales revenue**
- **Profit – Net profit from the order**
- **Quantity – Units sold**

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did interactivity improve the analysis experience in your dashboard?

Interactivity made the dashboard much easier to explore and understand. Instead of looking at static charts, I could filter the data by region, category, and date to focus on specific information. The drill-down feature helped analyze sales trends from yearly data down to monthly and daily levels. The parameter allowed switching between sales and profit without creating multiple charts. These interactive features made it easier to identify patterns, compare performance, and gain insights quickly. Overall, the dashboard became more user-friendly and supported faster decision-making based on the selected filters.

2. Which interactive feature (filters, drill-down, parameters, etc.) was most useful and why?

Among all the interactive features, I found the parameter to be the most useful because it allowed the same chart to display either Sales or Profit without creating separate visuals. This reduced dashboard clutter and made comparisons much easier. Other helpful features were:

- Filters/Slicers: Quickly focused the analysis on a specific region, category, or date range.
- Drill-down: Helped explore sales trends from yearly summaries to more detailed monthly and daily data.
- Tooltips: Displayed additional information without occupying extra dashboard space.

These features together made the dashboard more flexible and interactive.

3. What challenges did you face while implementing drill-down or dynamic filters?

The main challenge was understanding how Power BI handles hierarchies and field parameters. Initially, the parameter did not work because I used the Sales and Profit columns instead of creating DAX measures. I also had some difficulty finding the correct options for tooltips and drill-down because the interface in the latest Power BI version is different from many online tutorials. Another challenge was ensuring that all slicers filtered every visual correctly. After using Edit Interactions and creating the required measures, the dashboard behaved as expected and all interactive elements worked smoothly.

4. How do tooltips contribute to better dashboard storytelling?
Tooltips improve dashboard storytelling by providing additional details only when the user needs them. Instead of overcrowding the dashboard with too much information, they display important values when the user hovers over a visual. For example, while viewing sales by category, the tooltip can show total sales, profit, quantity, and region at the same time. This gives better context without making the dashboard look cluttered. Tooltips also make it easier to compare different data points and help users understand the insights more effectively.

5. What additional features would you add to make the dashboard more insightful?
To make the dashboard even more informative, I would consider adding the following features:

- A Top 10 Products chart to identify the best-performing products.
- Year-over-Year (YoY) sales and profit comparison to track business growth.
- Forecasting to predict future sales trends based on historical data.
- Conditional formatting to automatically highlight high and low-performing regions.
- A customer segmentation analysis to understand purchasing behavior.
- Additional KPI cards showing average order value and total number of orders.
- Bookmark buttons for quickly switching between different dashboard views.
- Custom report page tooltips with more detailed information for each category or regio

RESULT:

The retail sales dataset was successfully transformed, aggregated, and visualized into interactive dashboards in Tableau and Power BI. Filters, slicers, drill-downs, and parameters enabled dynamic exploration of sales and profit trends by region, product category, and time period. The resulting dashboards allowed for quick insights and deeper analysis, enhancing decision-making capabilities.

ASSESSMENT

Description	**Max Marks**	**Marks Awarded**
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	